

DES646 Mid-Term Assessment: Project Proposal

1 Project Title

AI-Powered Moodboard Generator: Enhancing Visual Ideation Through Generative Design

2 Team Details

Team: D-for-Design

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3 Project Aim

This project addresses time-intensive moodboard creation by developing an AI-powered web application that automatically generates personalized moodboards (color palettes, reference images, style suggestions) from natural language design briefs, reducing ideation time by 40-50%

4 Objectives

- Develop interactive Streamlit web tool transforming textual design briefs into visual moodboards with AI-generated color palettes and style-matched concepts
- Implement NLP keyword extraction and color theory algorithms mapping design requirements to visual elements across 8+ style categories
- Deploy working demo on Streamlit Cloud with comprehensive documentation for open-source contribution
- Prepare research paper for India HCI 2026 on AI-assisted moodboard generation's impact on designer productivity

5 Methodology / Technical Approach

AI/ML Methods: NLP keyword extraction (NLTK/spaCy) parsing design briefs and identifying mood descriptors. Color generation using rule-based algorithms and k-means clustering for palette creation across five mood categories (Warm, Cool, Neutral, Vibrant, Monochrome) . Framework designed for OpenAI DALL-E or Stable Diffusion integration; MVP uses curated placeholder system .

Tech Stack: Streamlit (web framework), Pillow (image processing), NumPy (computations), GitHub (version control), Streamlit Community Cloud (deployment) .

6 Expected Deliverables

- **Web Application:** Functional Streamlit app with text-to-moodboard generation, style selection, color palette visualization, user feedback mechanisms
- **Source Code:** GitHub repository with documented Python code, requirements.txt, README, deployment guides
- **User Study Results:** Quantitative data (time metrics, creativity scores) and qualitative feedback from 10-15 participants with statistical analysis
- **Technical Docs:** System architecture diagrams, API documentation, inline code comments
- **Video Demo:** 3-5 minute screen recording showcasing features and workflows

7 IP / Research Dissemination Plan

Publication Track (Primary): Targeting **India HCI 2026**, premier Indian HCI conference with tracks on creativity support tools and AI-augmented design . **Paper Title:** "AI-Assisted Moodboard Generation: Empirical Evaluation of Automated Visual Ideation Tools." **Contributions:** Empirical AI impact evaluation on design creativity; novel application addressing moodboard automation research gap ; open-source educational tool; human-AI creative workflow framework. **Co-authorship:** Instructor included per course policy.

Patent Track (Secondary): If unique algorithmic innovations emerge (semantic-to-visual mapping with adaptive learning), patent filing may be pursued. Innovation focus: lightweight architecture enabling deployment without expensive GPU infrastructure, democratizing AI design tools.

8 Resources Required

Software (Free): Python 3.8+, Streamlit, Pillow, NumPy, NLTK/spaCy, GitHub, Streamlit Community Cloud — **Hardware:** Standard laptop (8GB RAM), internet, no GPU needed — **Lab Access:** Computer lab (2-3 days testing), meeting space (1-2 days interviews) — **Optional (1000-1500):** OpenAI API credits, Unsplash/Pexels API — **Budget:** 0-2000 (zero mandatory budget)